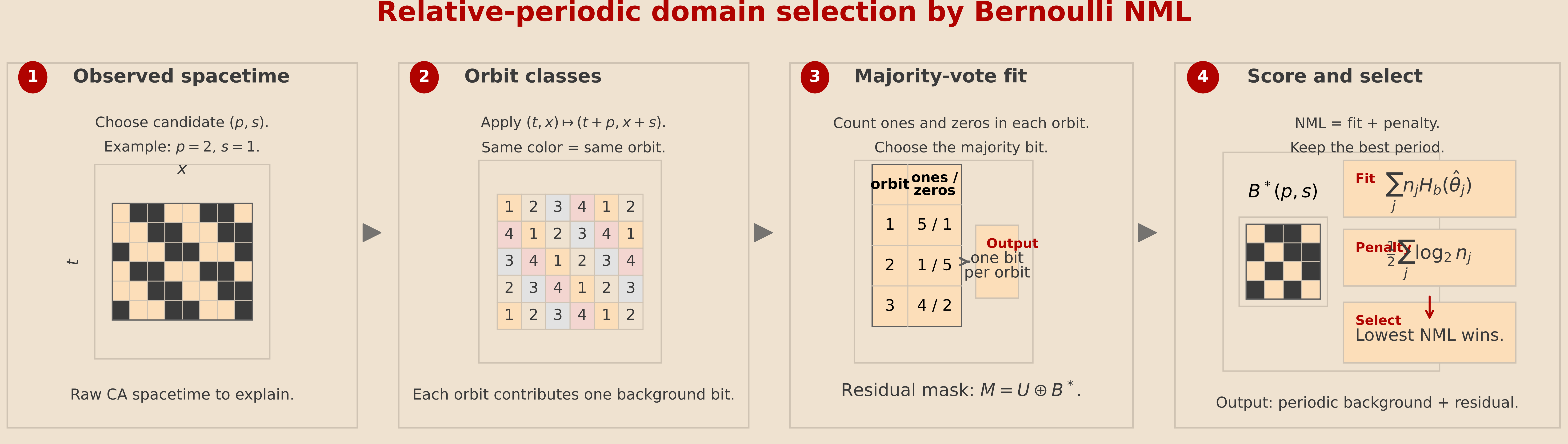
Relative-periodic domain selection by Bernoulli NML
1 Observed spacetime
Choose candidate (p, s).
Example: $p = 2$, $s = 1$.
x
t
Raw CA spacetime to explain.
2 Orbit classes
Apply $(t, x) \mapsto (t + p, x + s)$.
Same color = same orbit.
1 2 3 4 1 2
4 1 2 3 4 1
3 4 1 2 3 4
2 3 4 1 2 3
1 2 3 4 1 2
Each orbit contributes one background bit.
3 Majority-vote fit
Count ones and zeros in each orbit.
Choose the majority bit.
orbit
ones / zeros
1 5 / 1
2 1 / 5
3 4 / 2
Output
one bit per orbit
Residual mask: $M = U \oplus B^*$.
4 Score and select
NML = fit + penalty.
Keep the best period.
$B^*(p, s)$
Fit $\sum_j n_j H_b(\hat{\theta}_j)$
Penalty $\frac{1}{2}\sum_j \log_2 n_j$
Select
Lowest NML wins.
Output: periodic background + residual.